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Graduation Project

Prediction and Explainable AI-Based Clinical Decision Support System for Diabetic Nephropathy

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A graduation project submitted in partial fulfillment of the requirements for the degree of Bachelor of Computer Science and Information System

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Project No: F26_602 **Academic Year: 2025/ 2026**

Acknowledgment

With deep gratitude, we would like to thank God for giving us the strength, patience, and capability to do this project. We would also like to extend our deepest thanks to Culture and Science City and to the Dean of the Higher Institute of Computer Science and Information Systems for their complete cooperation and encouragement, which really helped us a lot in finishing this project. Their support and collaboration have been priceless; we are thankful for the opportunities they provided.

We would like to express our deepest gratitude to **Dr. Ahmed Gaber** and **Eng. Ream Salama** for their crucial support and guidance during our graduation project. Their dedication and expertise along with their encouragement have been of vital importance to our success. We feel privileged to study from persons as accomplished and committed as themselves. Their contributions as supervisors and mentors were vital to our success, and we will always be deeply grateful for their constant support.

Abstract

Diabetic nephropathy is one of the most serious problems that can happen to people suffering from diabetes. That is why people suffer from chronic kidney disease (CKD). It occurs slowly over time due to high blood sugar levels and damages the filters in the kidneys called nephrons as kidney damage may develop without noticeable symptoms. and cause structural function damage to the nephrons. If not predict early, it ends to renal failure either acute or chronic.

This project proposes the development of an innovative web-based Prediction and Clinical Decision Support System powered by Artificial Intelligence (AI). The system combines laboratory and follows up on the patient's condition. To guarantee uniform and evidence-based assessment, patient data are regularly examined and contrasted with accepted clinical standards for diabetic nephropathy.

The primary objective is to find and enhance early in (Stage I or II) and precise staging of chronic kidney disease (Stage I-V) based mostly on eGFR numbers Creatinine levels, HBA1C and others of lab tests. By producing clear, understandable results, the system enables doctors to grasp the main variables affecting risk categorization and disease progression using Explainable Artificial Intelligence (XAI) methods.

The suggested system seeks to predict and support clinical decision, enhance disease management efficiency, and help to slow the progression toward late renal failure by offering organized risk assessment, stage determination, and future positive progression prediction.

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Chapter 1: Introduction

1.1 Overview

More than 485 million people worldwide suffer from diabetic nephropathy which stands as one of the most frequent complications associated with diabetes. Diabetes represents the primary factor responsible for chronic kidney disease development whereas diabetic nephropathy leads to eventual kidney failure. The disease poses a threat because its early stage shows no signs which results in doctors diagnosing patients only after the condition has progressed to a severe state. The regular check-ups along with albumin to creatinine ratio and glomerular filtration rate tests serve as essential methods for detecting health issues at their initial stages.

Diabetic nephropathy occurs when patients with diabetes fail to control their blood sugar levels which leads to kidney glomeruli and tubule systems experiencing structural and functional damage. These changes lead to a steady decline of kidney functions. The kidneys progress toward their end stage when they lose their capacity to manage body fluids together with electrolytes which ultimately leads to the need for dialysis or kidney transplantation.

The clinical diagnosis of diabetic nephropathy is determined through a structured sequence of laboratory tests which use biomarkers for identification. The first and most basic measure of the kidney's filtration efficiency is **serum creatinine**. Then creatinine values provide the basis for calculating the estimated glomerular filtration rate which determines the chronic kidney disease stage.

Long-term glycemic control is assessed using **HbA1c**, as persistent elevation accelerates renal microvascular damage. Early kidney involvement is best detected using the **Urine Albumin to Creatinine Ratio (UACR)**, which identifies microalbuminuria before significant decline in **eGFR** occurs.

The Patients who do not receive treatment suffer from serious medical problems that lead to acute or chronic kidney failure.

1.2 Motivation

The timing of diabetic nephropathy development differs significantly between the two major types of diabetes. In Type 1 diabetes, screening for kidney complications typically begins 10-20 years after the initial diagnosis, as nephropathy often develops gradually over time. In contrast, in Type 2 diabetes, screening should be initiated immediately at diagnosis, since the disease may have been present for years without detection, allowing renal damage to progress silently.

The assessment process for diabetic nephropathy presents medical professionals with a significant obstacle because its evaluation requires multiple methods. The condition depends on the integration of multiple laboratory parameters and numerous risk factors including poor glycemic control (elevated HbA1c) and hypertension and smoking and obesity and dyslipidemia and prolonged disease duration. Doctors need to assess multiple biomarkers together with their clinical background to determine patient health status which includes laboratory testing. The analysis process which requires multiple dimensions creates a risk that doctors might miss early disease signs or fail to recognize initial disease symptoms or overlook clinical concerns.

1.3 Objective

- **System Overview**

The unified predictive model will utilize essential biomarkers, including:

- Serum Creatinine
- Estimated Glomerular Filtering Rate (eGFR)
- HbA1c
- Ratio of Urine Albumin to Creatinine (UACR)
- Urea
- uric acid

Furthermore, the system will link renal imaging data with laboratory results to give a more thorough analysis of kidney condition.

- **Explainable artificial intelligence (XAI)**

To ensure transparency and clinical trust, the system will integrate Explainable Artificial Intelligence (XAI) techniques.

This will let doctors/ patient:

- Know which laboratory variables most affect risk categorization.
- Understand how forecasts are produced.
- Evaluate the rate of chronic kidney disease (CKD) advancement more carefully.

1.4 Scope

The scope of the project is to:

- Early detection of diabetic nephropathy, particularly in Stage 1-2 before significant kidney function decline occurs.
- Analysis of major risk factors, including glycaemic control (HbA1c), hypertension, dyslipidaemia, obesity, and smoking.
- Utilization of essential laboratory investigations such as creatinine, eGFR, UACR, urea, and uric acid.
- Classification of chronic kidney disease (CKD) into its five clinical stages based on eGFR and albuminuria levels.
- Risk stratification for disease progression toward advanced CKD and kidney failure.
- Comparison of patient data with established clinical guidelines to enhance treatment alignment.
- Providing interpretable and transparent results using Explainable AI techniques to support medical decision-making.
- Development of a web-based platform to facilitate practical use by physicians in clinical settings.

The scope of the project does not replace medical diagnostic procedures.

1.5 General Constraints

There are several general limitations of this project, and those main limitations are:

- Limited availability of integrated and high-quality medical data
- Different methods of writing radiology reports between medical institutions
- The difficulty of obtaining large and medically tagged databases
- Ethical considerations and patient data privacy
- Adopting the system as a decision support system and diagnostic tool

1.6 Organization of the project

This book is organized as follows:

Chapter 1: Introduction Briefly describes the project, and its components and gives overviews.

Chapter 2: Theoretical Background and Tools Describes theoretical background.

Chapter 3: System Analysis discussed the architecture of our system.

Chapter 4: Design and Implementation Constraints

Chapter 2: Background and Previous work

2.1 Background

This chapter provides technological solutions that are powered by artificial intelligence and which are meant to diagnose, monitor, and manage diabetic nephropathy. The chapter examines the ways in which the higher order AI models and data modelling approaches can help to systematically process laboratory parameters and medical imaging data with the view of assisting clinical decision-making processes. This special focus is given to enhancing early-stage detection- especially at Stages 1 and 2 when intervention in the early stages of the disease can greatly slow down the disease progression.

In addition, this chapter discusses the integration about the build and incorporation of Explainable Artificial Intelligence (XAI) methods to provide transparency, interpretability, and clinical trustworthiness. Offering rational reasons to support predictive outcomes, these mechanisms should help to increase the confidence of physicians, ensure the ethical use of AI, and integrate it safely into real healthcare environments.

2.1.1 Pathophysiology

- Diabetic nephropathy (DN) [1] occurs through a complex interaction of the metabolic and hemodynamic mechanisms that are stimulated by longstanding hyperglycemia. One of the most primitive changes is glomerular hyperfiltration, an adaptive early response to the high levels of glucose with increases in intraglomerular pressure, which will eventually lead to structural damage.
- Progressive mesangial expansion, characterized by mesangial cell proliferation and excessive extracellular matrix deposition, leads to glomerulosclerosis and reduced filtration efficiency. Concurrently, thickening of the glomerular basement membrane (GBM) impairs selective permeability of the filtration barrier.
- In addition, any injury or loss of podocytes disrupts the integrity of the glomerular filtration barrier leading to the development of albuminuria - an early clinical sign of DN. Collectively, these pathologic changes lead to gradual decline of renal function, which eventually leads to chronic kidney disease (CKD), hypertension and eventually possibly end stage renal disease (ESRD).

- Early detection is important because Stages 1 or 2 do not produce any symptoms of the disease, even though microscopic damage is occurring. Delayed diagnosis can cause irreversible chart of loss nephrons and require renal replacement treatments, such as dialysis or kidney transplant.

2.1.2 Key Laboratory Tests in Diagnosing and Monitoring Diabetic Nephropathy

The clinical evaluation of DN relies heavily on laboratory investigations, which provide quantitative markers of kidney function, structural integrity, and systemic metabolic disturbances. These include: [2-3]

1. **Glycated Haemoglobin (HbA1c)**

HbA1c reflects average blood glucose levels over the preceding 2-3 months. Poor glycaemic control, indicated by elevated HbA1c, is strongly correlated with accelerated nephropathy progression.

Normal Range:

- **Standard (not diabetic):** < 5.7%
- **Pre-diabetes:** 5.7%-6.4%
- **Diabetes:** at least 6.5%
- Suggested target in the majority of diabetic patients: <7% (may differ personally)

Clinical Risk in Diabetic Nephropathy:

- HbA1c > 7% suggests bad glycaemic control.
- HbA1c > 8-9% correlates greatly with sped glomerular damage.

Repeated hyperglycaemia results in:

- glomerular hyperfiltration
- thickening of the basement membrane
- Mesangial inflation
- Gradual decrease in eGFR

2. Serum Creatinine

A **creatinine lab test** measures the level of creatinine a waste product from muscle metabolism in your blood or urine to assess kidney function. High blood creatinine levels may indicate impaired kidney function, while low levels can be linked to reduced muscle mass or malnutrition.

Key Points:

Purpose: The test helps evaluate how well two kidneys filter waste from your blood. It's commonly included in basic metabolic panels (BMP) or comprehensive metabolic panels (CMP) during routine checkups.

- **Normal Ranges** (blood creatinine):
- **Adult males**: 0.7-1.2 mg/dL
- **Adult females**: 0.5-1.0 mg/dL

Note: Ranges vary slightly by lab and individual factors like age, sex, and muscle mass.

- **Interpretation**:
  - **High creatinine**: May signal kidney disease, dehydration, muscle injury (rhabdomyolysis), or conditions like heart failure or diabetes.
  - **Low creatinine**: Can result from low muscle mass, malnutrition, or severe liver disease.
- **Testing Process**:
  - **Blood test**: A healthcare provider draws blood from a vein (usually takes <5 minutes).
  - **Urine test**: Often requires a 24-hour urine collection to measure creatinine clearance, which helps estimate glomerular filtration rate (eGFR) a more accurate measure of kidney function.
- **Important Notes:**
  - A single abnormal result doesn't diagnose kidney disease repeat testing and additional assessments (like eGFR or urine albumin-to-creatinine ratio) are often needed.
  - **Creatinine alone is not sufficient** to diagnose kidney issues. It's used in combination with other tests to determine kidney health.

3. **Estimated Glomerular Filtration Rate (eGFR)**
   - The eGFR is a key measure of kidney function. It estimates how much blood the kidneys filter per minute and is adjusted for body surface area. It is the primary marker used to stage chronic kidney disease (CKD) and to monitor the progression of diabetic nephropathy**.**

   - Calculated from serum creatinine, age, sex, and sometimes race, is a widely accepted measure for CKD staging:

CKD Stage	eGFR (mL/min/1.73 m²)	Clinical Meaning
Stage 1	≥ 90	Kidney damage present but filtration is normal. Often asymptomatic.
Stage 2	60-89	Mild reduction in kidney function. May have microalbuminuria.
Stage 3a	45–59	Moderate decrease in kidney function. Symptoms may appear (fatigue, mild oedema).
Stage 3b	30-44	Moderate-severe decrease in function. Increased risk of complications.
Stage 4	15-29	Severe decrease in kidney function. Symptoms more evident (oedema, hypertension, anaemia).
Stage 5 (ESRD)	< 15	Kidney failure (end-stage renal disease). Dialysis or transplantation required.

eGFR offers a dynamic view of kidney function, aiding clinicians in monitoring disease progression.

4. **Urine Albumin-to-Creatinine Ratio (UACR)**

UACR is the most sensitive marker for early DN detection. It quantifies urinary albumin excretion relative to creatinine, reflecting glomerular permeability:

- **Normal:** <30 mg/g
- **Microalbuminuria**: 30-300 mg/g
- **Macroalbuminuria:** >300 mg/g

Microalbuminuria often precedes detectable changes in serum creatinine, highlighting its value in early-stage diagnosis.

5. Blood Urea Nitrogen (BUN)

BUN rises in advanced kidney dysfunction and is used in conjunction with creatinine to assess renal excretory function.

Usual range: 7-20 mg/dL

- **Risk Interpretation:**
  - Moderate rise (20-30 mg/dL) could indicate dehydration or early kidney dysfunction.
  - Moderate elevation (30-50 mg/dL) lowers renal clearance.
  - Severe elevation (over 50 mg/dL) causes late kidney disease.
- Alongside serum creatinine, BUN is viewed as follows:
  - High BUN and creatinine cause decreased kidney filtration.
  - High BUN on its own might point to dehydration.

Note: High BUN implies a buildup of nitrogenous garbage brought on by decreasing eGFR.

6. Uric Acid

Hyperuricemia has been associated with inflammation, endothelial dysfunction, and worsening renal outcomes in DN patients.

- **Usual Range:**
  - Men: 3.4 to 7.0 mg/dL
  - Women: 2.4-6.0 mg/dL
- **Risk Levels:**
  - 7 mg/dL results Hyperuricemia
  - 8-9 mg/dL results in higher risk for endothelial and inflammation.

- **Clinical significance in DN:**
  Hyperuricemia links with:
  - Greater oxidative stress
  - Endothelial damage
  - More rapid CKD development
  - Increased cardiovascular risk

7. **Urine Analysis**

Routine urinalysis can identify glucosuria, haematuria, and urinary casts, offering additional clues to renal tubular and glomerular dysfunction.

- **Glucosuria**
  - Normally absent.
  - Presence with normal blood glucose → possible renal tubular dysfunction.
  - Presence with high glucose → poor glycaemic control.
- **Haematuria**
  - RBCs per high power field → abnormal.
  - May suggest glomerular damage.
- **Urinary Casts**
  - Hyaline casts → may be normal.
  - RBC casts → glomerulonephritis.
  - Granular casts → tubular injury.

UACR (mg/g)	**Interpretation**
<30	Normal
30–300	Microalbuminuria (Early DN)
>300	Macroalbuminuria (Advanced DN)

2.1.3 Risk Factors and Causes

In late-stage or neglected DN in patients (Stages 4-5), the following may develop due to adverse systemic fluid overload and uremia: [4]

- Generalized oedema (fluid in legs)
- Periorbital oedema (eyelid swelling)
- Pulmonary oedema (fluid in lungs)
- Shortness of breath

- Mental disturbances such as uremic encephalopathy
- Coma

These manifestations are important for early diagnosis and intervention before the occurrence of acute or chronic renal failure.

2.1.4 Relevance to the Proposed System

Despite the availability of multiple laboratory and clinical markers, early detection remains challenging due to: [5]

- Normal creatinine levels in early-stage
- Subtle interactions between multiple biomarkers that are difficult for clinicians to interpret manually

Integrating these laboratory parameters Creatinine, eGFR, UACR, HbA1c, Uric Acid and others into an AI-based Clinical Decision Support System (CDSS) enables:

- Early-stage detection (Stages 1-2)
- Risk prediction for disease progression
- Automated CKD staging
- Identification of complex patterns across multi-parametric lab data
- Explainable outputs to support clinical decision-making

2.2 Previous Work

2.2.1 Application of AI in Diabetic Nephropathy

Recent research has leveraged Machine Learning (ML) algorithms to predict the onset and progression of DN, primarily using clinical and laboratory datasets. Commonly applied algorithms include:

- ➢ Random Forest (RF) – robust to overfitting and effective in multi-feature datasets
- ➢ Support Vector Machines (SVM) – useful for classification of early vs. late DN stages
- ➢ XGBoost – gradient boosting framework offering high predictive accuracy

Common input items are: HbA1c, serum creatinine, eGFR, albuminuria and blood pressure. Although these models can be quite accurate, they are often referred to as black-boxes in that they give prediction without giving the logic behind the decision.

2.2.2 AI in Chronic Kidney Disease Staging

The AI-based methods have been used in staging CKD, predicting its progression, and identifying high-risk patients as well. Nevertheless, a number of limitations are noticed:

- Most of the models use eGFR as the main factor and disregard the entire range of biochemical markers.
- Extensive panel integration of the laboratory is lacking.
- There are normally no explainable AI mechanisms that may help clarify model decisions to clinicians.

2.2.3 Clinical Decision Support Systems (CDSS)

Traditional CDSS are designed to provide:

- Alerting for abnormal lab values
- Medication dosage recommendations
- Basic risk scores

Limitations of traditional CDSS include:

- Dependence on static, rule-based logic
- Lack of predictive learning capabilities
- Minimal personalization to individual patient data

Modern AI-enhanced CDSS aim to address these gaps by integrating predictive analytics, multi-parametric data, and interpretability.

2.2.4 Research Gap

Although the AI use in nephrology has made a great advancement, the existing Diabetic Nephropathy (DN) predictive systems continue to have severe methodological and clinical limitations that limit their feasibility in practice. [6]

The primary gaps in the research may be outlined as follows:

1. **Reduced Multi-biomarker Integration**
    - Most existing models rely primarily on traditional renal indicators such as creatinine, eGFR, HbA1c, and albuminuria. However, diabetic nephropathy is a multifactorial and systemic disease.
    - In many cases, especially for patients with rapidly changing kidney function, daily or every-other-day lab tests are required to monitor key biomarkers (e.g., creatinine, eGFR, UACR, HbA1c, electrolytes).
    - This limited integration reduces the ability of models to capture the complex pathophysiological interactions influencing disease progression.
2. **Weak Early-Stage Diagnosis (Stage 1-2)**
    - The main issue is that they fail to detect the disease in its early stages (Stage 1-2), which is the period when therapeutic interventions are most effective in preventing kidney deterioration.
    - As a result, these systems often identify the disease too late, when kidney function has already significantly declined (Stages 3-5) and the patient is already at risk of complications such as hyperkalemia, fluid overload, or chronic kidney failure.
    - Additionally, Clinicians do not effectively track disease progression over time, and therefore fail to provide early warnings for rapid progressors.
3. **Lack of Explainability and Clinical Transparency**
   Many high-performance AI models act as systems without a clear explanation. The absence of feature significance analysis or understandable reasoning limits physician confidence and reduces real-world clinical adoption. In medical decision-making, transparency is essential for safe and ethical implementation.
4. **Limited Integration with Clinical Workflow**
   A large proportion of proposed systems remain research prototypes rather than fully operational Clinical Decision Support Systems (CDSS). They often lack real-time dashboards, automated CKD staging, guideline comparison, and actionable clinical alerts, reducing their practical value in daily clinical practice.

2.2.5 Contribution of the Proposed System

Building upon the previously identified research gaps, the proposed system introduces a comprehensive, explainable, and clinically integrated AI-based Clinical Decision Support System (CDSS) specifically designed for Diabetic Nephropathy (DN). The system directly addresses the limitations of existing approaches through the following targeted contributions:

1. **Development Multi-Biomarker Integration**
   Unlike traditional models depending on restricted renal indices, the suggested system incorporates a wider spectrum of laboratory criteria including creatinine, eGFR, HbA1c, UACR, urea, electrolytes, uric acid, and mineral metabolism indicators. This method sharpens the capacity and early predicts to record the multifactorial character of DN.
2. **Enhanced Early-Stage Detection (Stage 1-2)**
   Early-stage diabetic nephropathy (Stage 1-2) is especially emphasized in the system, hence enabling quick treatment before major reduction in kidney function develops. It also enables risk stratification to find people with a greater chance of quick advancement.
3. **Lack of Explainability**
   The system uses Explainable AI (XAI) methods to address the interpretability limitation, and they include:
   - Feature importance ranking
   - Local prediction explanations.

   Summaries of reasoning that can be clinically interpreted.
   This will guarantee transparency, physician trust, and will be in line with the ethical and regulatory requirements of medical AI systems.

4. **Integrated Clinical Decision Support**
   - Rather than a stand-alone predictive model, the suggested system is an integrated Clinical Decision Support System (CDSS). It offers:
     - Automatic CKD grading (Stage I-V) depending on eGFR and albuminuria
     - Categorization of patient risk (low / moderate / high)
     - Real-time clinical monitoring dashboards.
     - Actionable, interpretable outputs bridging algorithmic prediction and clinical decision-making.

- Furthermore, to overcome the constraint of temporal analysis in current research, the system includes longitudinal surveillance and basic prognosis modelling. It lets one:
  - Regular tracking of eGFR patterns over time
  - Prediction of illness advancement risk
  - Early detection of fast progressors

The system improves proactive patient management and enables timely therapeutic intervention without substituting doctor judgment by combining dynamic trend analysis with organized clinical outcomes.

Chapter 3: planning and Analysis

3.1 Planning

Planning represents the foundational phase of the Software Development Life Cycle (SDLC). For healthcare-based artificial intelligence systems, planning extends beyond technical considerations to include ethical, operational, financial, and regulatory dimensions. The proposed Prediction and Explainable AI-Based Clinical Decision Support System (CDSS) for Diabetic Nephropathy requires structured planning to ensure clinical reliability, patient data security, and long-term sustainability.

The planning phase includes feasibility analysis, cost estimation, infrastructure planning, risk management, stakeholder coordination, and timeline structuring. Because the system directly influences medical decision-making, comprehensive planning minimizes potential clinical risks and implementation challenges.

3.1.1 Feasibility Study and Estimated Cost

3.1.1.1 Technical Feasibility

Technical feasibility evaluates the availability of required technologies, infrastructure, and expertise. The system leverages mature machine learning frameworks such as Scikit-learn and XGBoost for predictive modeling. Explainability is implemented using SHAP to provide transparent feature contribution analysis.

The architecture follows a multi-layer design consisting of presentation, application, AI processing, and database layers. Backend implementation will utilize Django or Flask frameworks, while MySQL will manage structured patient data. Frontend components will be developed using HTML, CSS, JavaScript, Bootstrap, React to ensure cross-platform compatibility.

Hardware requirements include a cloud-based server with a minimum of 16GB RAM, encrypted communication channels via SSL certification, secure authentication mechanisms, and automated backup procedures. Given the availability of these technologies and the team's expertise, the project is technically feasible.

3.1.1.2 Economic Feasibility

Economic feasibility assesses whether the projected financial investment is justified. Since the project utilizes open-source technologies, licensing costs are eliminated. The primary expenses relate to hosting infrastructure, server configuration, and maintenance.

Cost Component	Estimated Cost (USD)
Development Tools	0 (Open Source)
Cloud Hosting (1 Year)	\$300 – \$500
Server Infrastructure	\$800 – \$1200
Security & SSL Certificates	\$100
Maintenance & Updates	\$300

The total estimated cost ranges between \$1500 and \$2000. Early detection of diabetic nephropathy can significantly reduce dialysis costs and long-term complications, thereby justifying the investment.

3.1.1.3 Operational Feasibility

Operational feasibility examines how effectively the system integrates into existing clinical workflows. The system uses laboratory parameters already familiar to physicians, such as HbA1c, eGFR, creatinine, and UACR. By automating interpretation rather than introducing new metrics, workflow disruption is minimized.

3.1.1.4 Legal and Ethical Feasibility

The system complies with healthcare data protection standards by implementing encrypted communication, role-based authentication, and audit logging. Explainable AI mechanisms ensure transparency and prevent black-box decision-making, thereby promoting ethical AI usage.

3.1.2 Gantt Chart

The project timeline spans approximately six months and is divided into sequential development phases with clearly defined objectives and deliverables.

Phase	Duration
Literature Review and Research	3 Weeks
Requirements Analysis	2 Weeks
System Design and UML Modeling	3 Weeks
Dataset Collection and Preprocessing	4 Weeks
Model Development and Evaluation	4 Weeks
Explainable AI Integration	2 Weeks
Web Development and Integration	4 Weeks
System Testing and Validation	3 Weeks
Documentation and Final Review	3 Weeks

3.2 Analysis and Limitations of Existing System

Existing diabetic nephropathy detection approaches rely primarily on manual interpretation or static rule-based CDSS frameworks. These systems fail to capture nonlinear relationships between multiple biomarkers and lack interpretability mechanisms.

- Limited multi-biomarker integration.
- Weak early-stage detection.
- Absence of explainable AI.
- No longitudinal monitoring.
- Limited personalization.

3.3 Need for New System

The global rise in diabetes prevalence necessitates advanced predictive tools capable of early detection of diabetic nephropathy. Artificial intelligence enables detection of complex biomarker interactions, enhancing early intervention and improving patient outcomes.

3.4 Analysis of System

3.2.1 User Requirements

- Secure authentication system.
- Structured laboratory data entry interface.
- Automatic CKD stage computation.
- Risk classification (Low/Moderate/High).
- Explainable AI output.
- Trend visualization and report generation.

3.2.2 System Requirements

- Cloud-based infrastructure.
- Python AI environment.
- Django/Flask backend.
- PostgreSQL database.
- Browser compatibility.

3.2.3 Domain Requirements

- CKD staging aligned with eGFR thresholds.
- Albuminuria classification using UACR.
- HbA1c interpretation according to guidelines.

3.2.4 Functional Requirements

1. User authentication.
2. Laboratory input validation.
3. CKD stage calculation.
4. AI risk prediction.
5. Explainable feature importance.
6. Patient history tracking.
7. Report generation.

3.2.5 Non-Functional Requirements

- Performance under 5 seconds.
- Encrypted data transmission.
- High availability.
- Scalable architecture.

3.2.6 Advantages of the New System

- Early-stage CKD detection.
- Multi-biomarker integration.
- Explainable AI transparency.
- Improved clinical workflow.

3.2.7 User Characteristics

- **Physicians:** primary users, medical expertise, limited AI knowledge focus on interpretability
- **Administrators:** manage technical aspects
- **Patients:** indirect benefits through early detection and monitoring

Chapter 4: Design and Implementation Constraints

4.1 Design and Implementation Constraints

There are several technical, clinical, and operational limitations in the design and execution of the suggested AI-based Clinical Decision Support System (CDSS) for Diabetic Nephropathy.

- **Data Availability and Quality**
  In addition to renal imaging results, the system depends much on ordered laboratory data like creatinine, eGFR, HbA1c, UACR, urea, and uric acid. Model accuracy and generalizability may be impacted by laboratory standard fluctuations, missing data, erratic reporting formats, or constrained dataset size.
- **Clinical Guideline Variability**
  Patient information is matched with known medical guidelines (such as CKD staging criteria). Clinical guidelines may change from time to time, but suggestions can vary somewhat across different international organizations. Thus, the system has to be set up such that adaptable changes may be made without significant physical changes.
- **Model Interpretability Requirement**
  This project calls for the inclusion of Explainable Artificial Intelligence (XAI), unlike only predictive systems. Unless they can deliver interpretable outcomes fit for clinical use, this restriction limits the usage of some very complicated "black-box" models.
- **Ethical and Privacy Limitations**
  Handling medical information calls for exact adherence to patient privacy and data protection norms. Secure data storage, encrypted transfer, and limited access to sensitive information are all conditions the system has to meet.
- **Technical Infrastructure Constraints**
  Standard clinical systems and devices must all be compatible with the web-based platform. Deployment may be hampered by constraints in internet connectivity, computational resources, or interface with already in place hospital information systems.

4.2 Assumptions and Dependencies

- **Assumptions**

The project is developed under the following assumptions:

- Laboratory results entered into the system are accurate and validated.
- Standardized equations are used to compute eGFR creatinine, eGFR, HbA1c, UACR, urea, and uric acid values (e.g., CKD-EPI formula).
- Physicians will employ the system as a decision-support tool rather than as a replacement for their own medical expertise.
- Patients in the database are diagnosed with diabetes (Type 1 or Type 2); hence, screening procedures are observed appropriately.

- **Dependencies**

The successful operation of the system depends on:

- Availability of dependable and reasonably large medical databases for validating and training the artificial intelligence system.
- Revised clinical recommendations for diabetic nephropathy and CKD staging.
- Working with medical experts for validation and usability input.
- Stable web hosting and a safe server infrastructure.
- Continuous system maintenance and regular model training help to preserve performance accuracy.

4.3 Risk and Risk Management

1. **Technical Risks**
   - ***Risk:*** Wrong predictions caused by low quality or biased training data and Overfitting of the AI model.
   - ***Mitigation:*** applies cross-validation procedure on diverse datasets and continuously evaluates model performance and Apply regularization techniques, appropriate train-test partitioning and External validation.

2. **Clinical Risks**
    - ***Risk:*** Physician overreliance on the system and poor interpretation of the outputs of AI by the user.
    - ***Mitigation:*** Be able to explain clearly using XAI techniques, formatted visual output, and other guidance notes to the interface and make it clear to the interface users that the system is a support tool and not a substitute of clinical expertise
3. **Data and Security Risks**
    - ***Risk:*** Breach of data or unauthorized access.
    - ***Mitigation:*** Use encryption standards, secure authentication, and adherence to data protection standards of healthcare.
4. **Operational Risks**
    - ***Risk:*** Opposition to adoption by healthcare professionals and medical guideline alteration.
    - ***Mitigation:*** Develop a friendly interface, provide training and show clinical validation outcomes and Design modular architecture, which allows easy clinical rule modification.

4.4 System Models

4.4.1 Classical Analysis

4.4.1.1 Context Diagram

It is a high-level view of a system. It provides a basic sketch to define an entity based on its scope, boundaries, and relation to external components like stakeholders as shown in Figure 4.1.

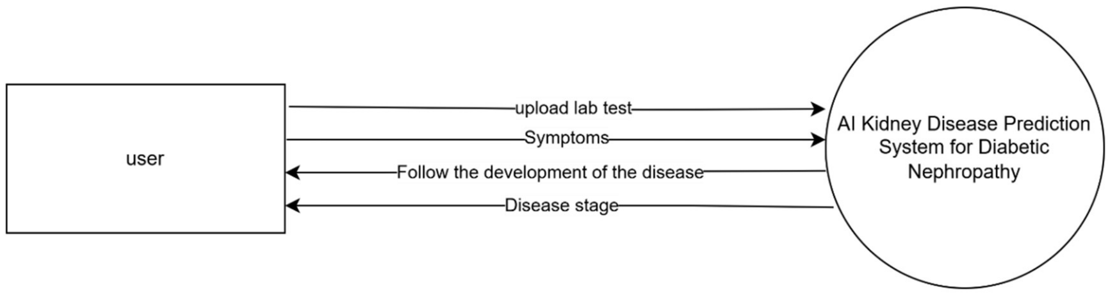

Figure 1: Context Diagram

4.4.2 UML Diagrams

4.4.2.1 Use-Case Diagram

A use case: A use case is a description of a specific scenario or situation in which a system or product is used to achieve a particular goal or outcome. It outlines the various steps, actions, and interactions between the user and the system as shown in Figure 4.2.

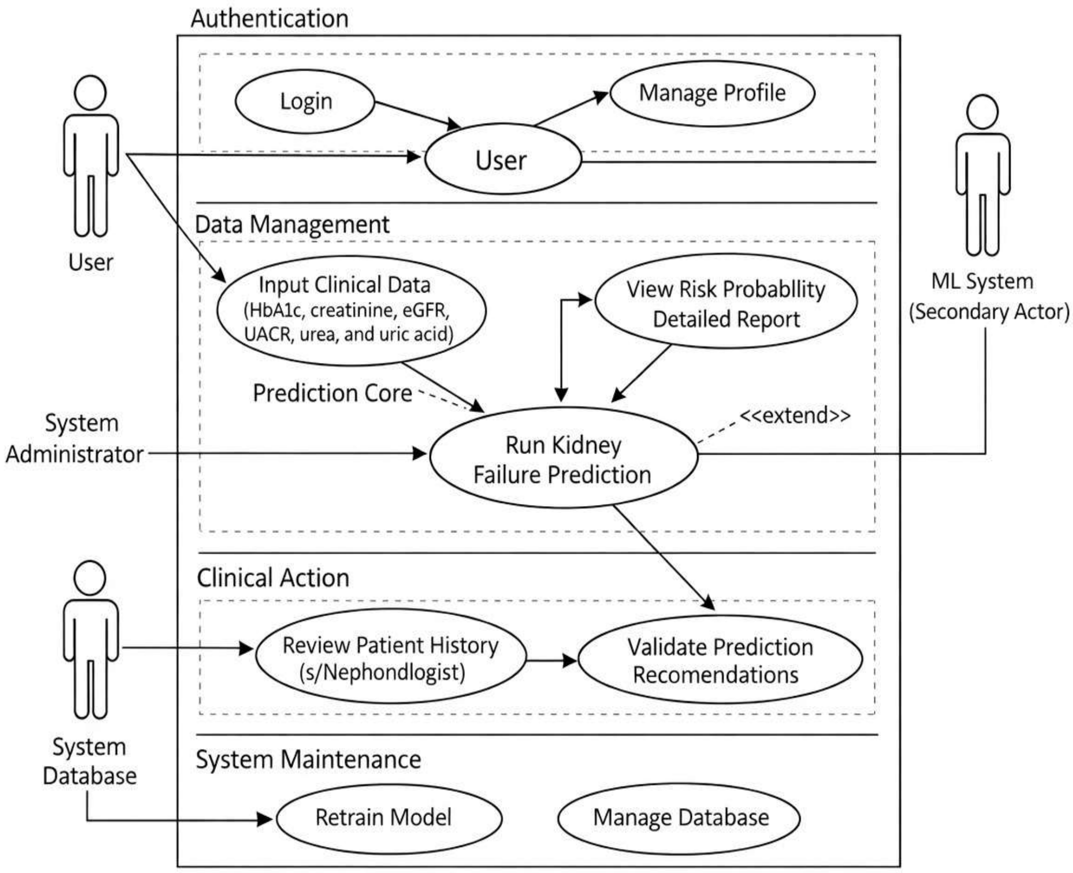

Figure ٢ : Use-case Diagram

4.4.2.2 Sequence Diagram

The sequence diagram is an interaction diagram that shows how processes operate with one another and in what order. A sequence diagram shows object interactions arranged in a time sequence. It depicts the objects and classes involved in the scenario and the sequence of messages exchanged between the objects needed to carry out the functionality of the scenario as shown in Figure 4.2 and Figure 4.3.

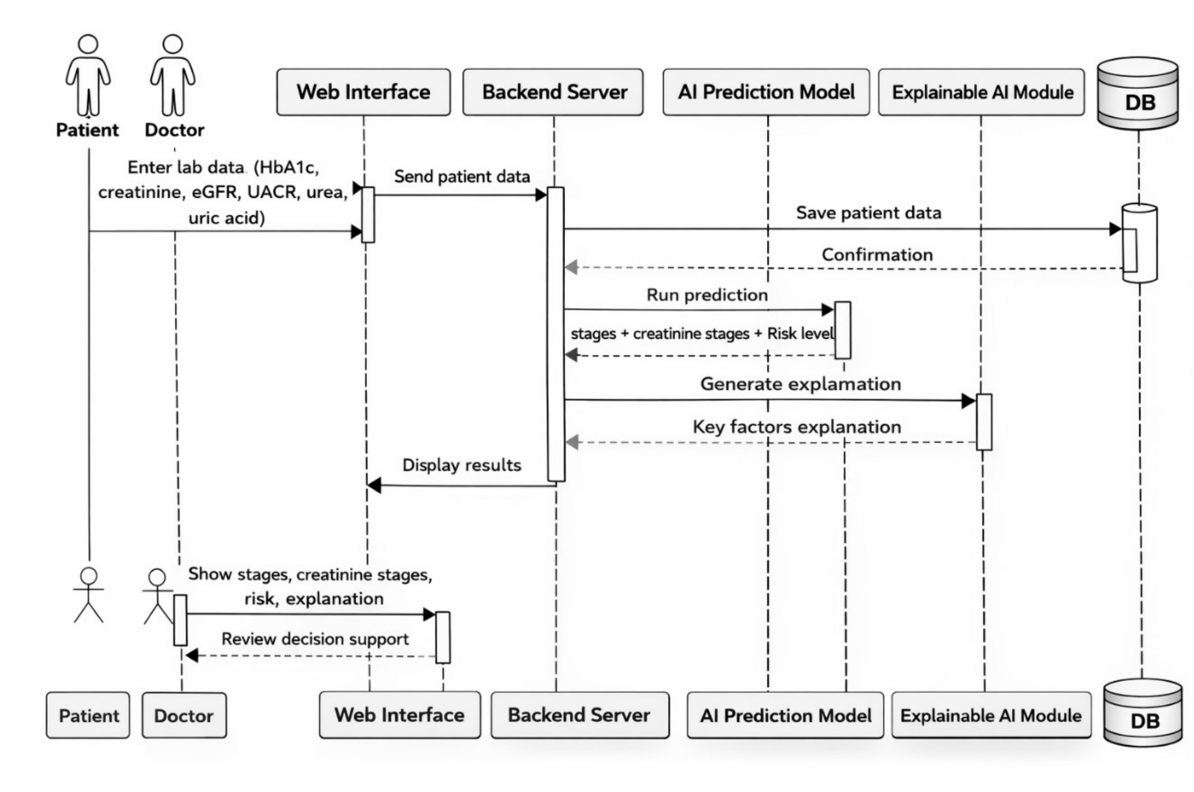

Figure 3: Sequence Diagram

4.4.2.3 Activity Diagram

An activity diagram is a chart that captures the dynamic behavior of the system and shows the message flow from one activity to another as shown in Figure 4.4

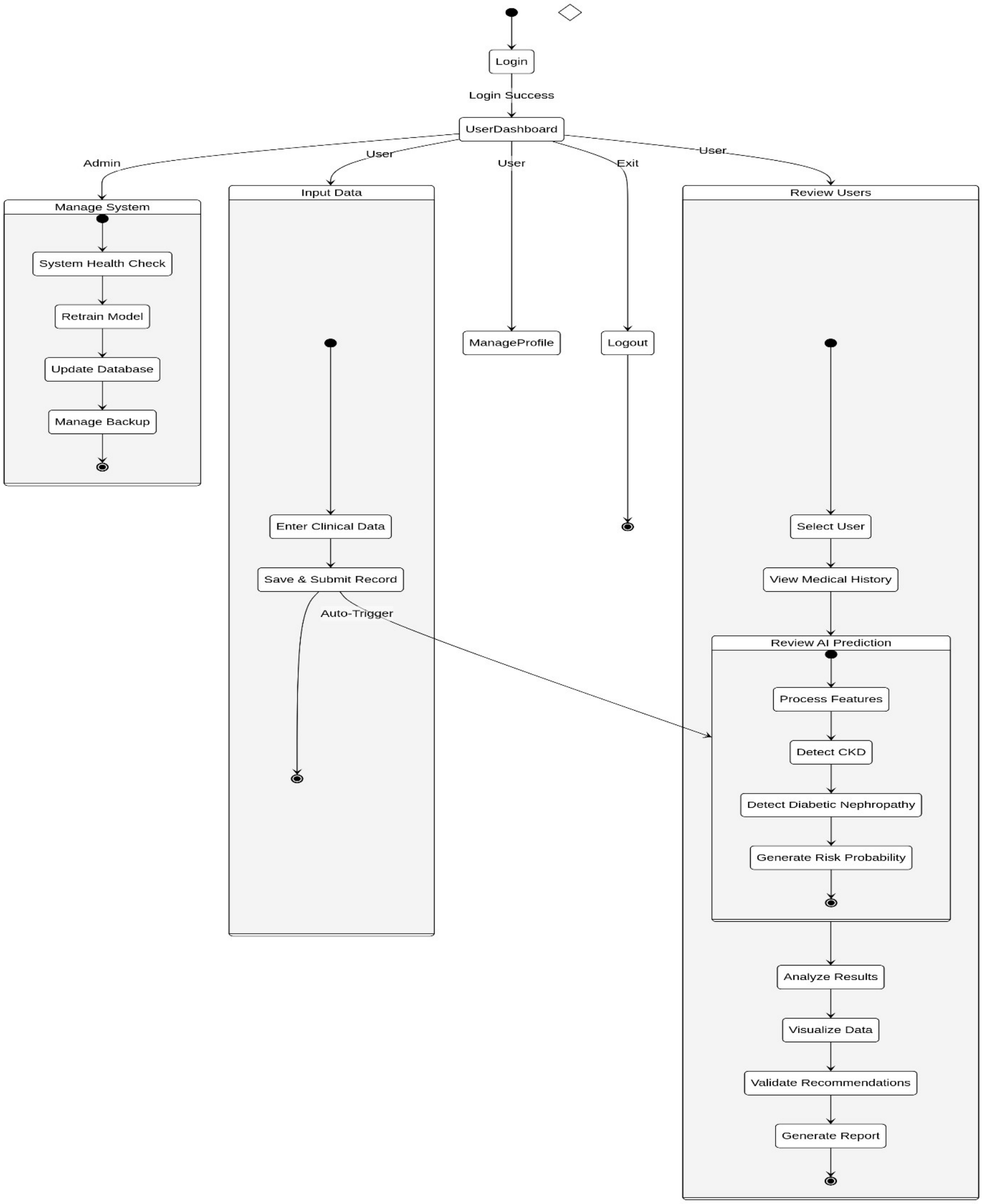

Figure ٤ : Activity Diagram

4.4.2.4 Class Diagram

It is an illustration of the relationships and source code dependencies among classes in the Unified Modeling Language (UML) as shown in Figure 4.5.

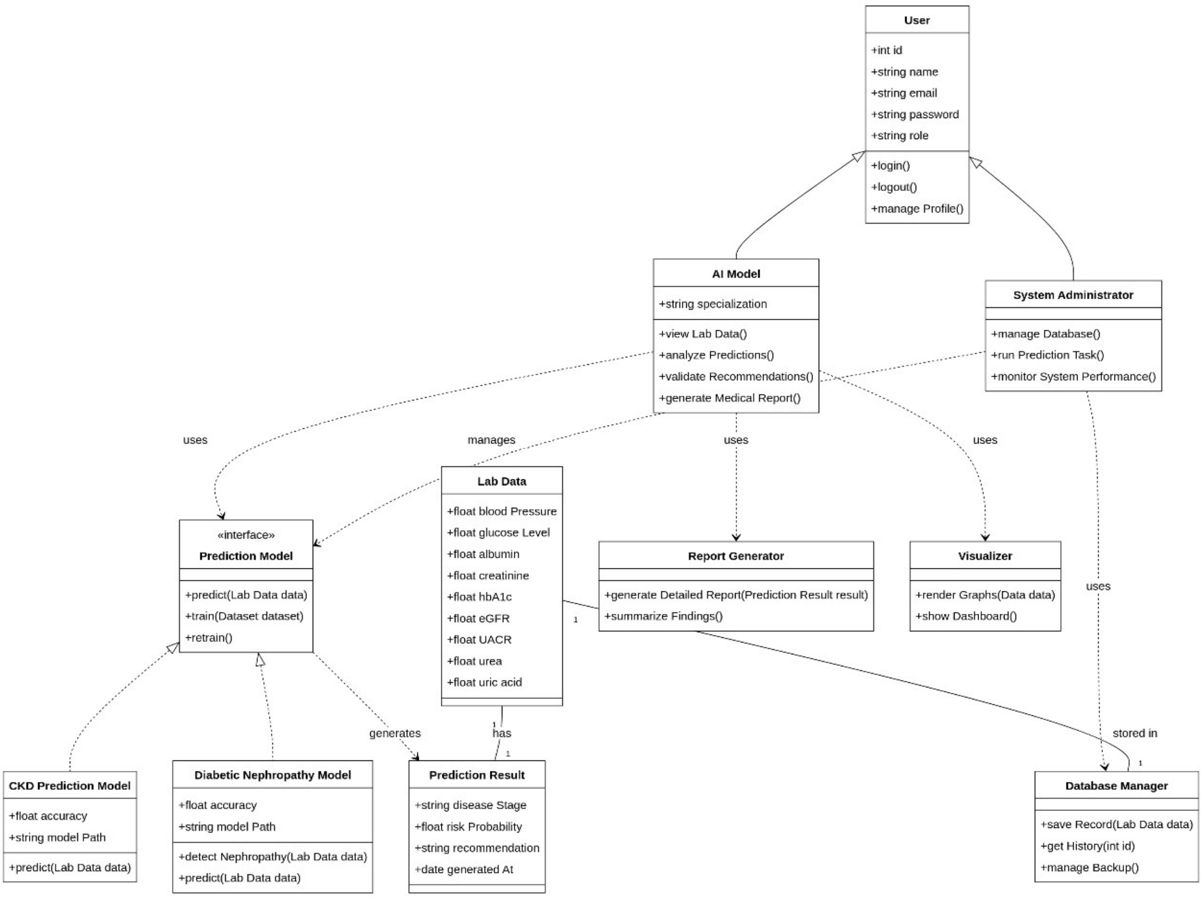

Figure ∘ : Class Diagram

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Link: Diabetic nephropathy

ملخص المشروع

يعد اعتلال الكلية السكري من أخطر المشاكل التي يمكن أن تحدث للأشخاص الذين يعانون من مرض السكري. هذا هو السبب في أن الناس يعانون من أمراض الكلى المزمنة (كد). يحدث ببطء مع مرور الوقت بسبب ارتفاع مستويات السكر في الدم ويتلف المرشحات في الكلى تسمى النيفرون حيث قد يتطور تلف الكلى دون أعراض ملحوظة. وتسبب تلف الوظيفة الهيكلية للنيفرون. إذا لم يكن التنبؤ في وقت مبكر ، فإنه ينتهي إلى الفشل الكلوي إما حادة أو مزمنة.

يقترح هذا المشروع تطوير نظام مبتكر للتنبؤ على شبكة الإنترنت ودعم القرار السريري مدعوم بالذكاء الاصطناعي. يجمع النظام بين المختبر ويتابع حالة المريض. لضمان التقييم الموحد والقائم على الأدلة ، يتم فحص بيانات المريض بانتظام وتتناقض مع المعايير السريرية المقبولة لاعتلال الكلية السكري.

الهدف الأساسي هو إيجاد وتعزيز في وقت مبكر (المرحلة الأولى أو الثانية) والتدريج الدقيق لمرض الكلى المزمن (المرحلة الأولى إلى الخامسة) يعتمد في الغالب على أرقام إغفر مستويات الكرياتينين ، هبا ١ ج وغيرها من الاختبارات المعملية. من خلال إنتاج نتائج واضحة ومفهومة ، يمكن النظام الأطباء من فهم المتغيرات الرئيسية التي تؤثر على تصنيف المخاطر وتطور المرض باستخدام أساليب الذكاء الاصطناعي القابلة للتفسير.

يسعى النظام المقترح إلى التنبؤ بالقرار السريري ودعمه ، وتعزيز كفاءة إدارة المرض ، والمساعدة في إبطاء التقدم نحو الفشل الكلوي المتأخر من خلال تقديم تقييم منظم للمخاطر ، وتحديد المرحلة ، والتنبؤ بالتقدم الإيجابي في المستقبل.